

## EXPERIMENT-19 STUDENT-TEACHER-SUB-CODE

### Aim

To create a Prolog database representing the relationship between **students, teachers and subject codes**.

### Objectives

- To represent relationships using facts.
- To retrieve teacher and subject information.
- To understand logical queries in Prolog.

### Algorithm

1. Define student, teacher and subject-code facts.
2. Create relationships between students, teachers and subjects.
3. Query the database.
4. Prolog matches the facts.
5. Display the result.

### Flowchart

START

|

v

Enter Student,

Teacher and Subject

|

v

Store facts

|

v

Enter query

|

v

Search matching facts

|

v

Display result

|

v

STOP

## Prolog Program

% Student-Teacher-Subject Code Database

student\_teacher\_subject(harshini, ramesh, cs101).

student\_teacher\_subject(priya, kumar, cs102).

student\_teacher\_subject(rahul, ramesh, cs103).

student\_teacher\_subject(anil, kumar, cs104).

student\_teacher\_subject(sneha, ramesh, cs105).

## Query

?- student\_teacher\_subject(harshini, Teacher, Code).

## Output

```
i:- initialization(main).
% Student-Teacher-Subject Database
student_teacher_subject(harshini, ramesh, cs101).
student_teacher_subject(priya, kumar, cs102).
student_teacher_subject(rahul, ramesh, cs103).
student_teacher_subject(anil, kumar, cs104).
student_teacher_subject(sneha, ramesh, cs105).

% Display all records
main :-
    student_teacher_subject(Student, Teacher, Subject),
    write(Student),
    write(' - '),
    write(Teacher),
    write(' - '),
    write(Subject),
    nl,
    fail.
main.

i:- initialization(main).
```

STDIN  
Input for the program ( optional )

Output:  
harshini - ramesh - cs101  
priya - kumar - cs102  
rahul - ramesh - cs103  
anil - kumar - cs104  
sneha - ramesh - cs105  
harshini - ramesh - cs101  
priya - kumar - cs102  
rahul - ramesh - cs103  
anil - kumar - cs104  
sneha - ramesh - cs105

## **Result**

The Student-Teacher-Subject Code relationship was successfully represented and queried.

## **Conclusion**

Thus, Prolog was successfully used to represent and retrieve relationships among students, teachers and subject codes.